



**SECP2523 DATABASE (WBL)**  
**SECTION 02**  
**ASSIGNMENT 1**

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## Assignment 2 SECPH 2523

### QUESTION 1

Table 2.1 STUDENTFINE records data on traffic offences by students in a college.

| StudentID | Name      | LicenseNo | CarReg  | ReceiptNo | Date     | OffenceCode | Fine(RM) |
|-----------|-----------|-----------|---------|-----------|----------|-------------|----------|
| F1011     | CK Fong   | P100      | JGK9201 | 00-345    | 10/03/06 | 2           | 30.00    |
|           |           |           |         | 00-690    | 15/03/06 | 1           | 20.00    |
|           |           |           |         | 00-810    | 20/03/06 | 2           | 30.00    |
| L782      | AdamJames | P290      | JJL99   | 00-459    | 13/03/06 | 1           | 20.00    |
| H762      | MalaDewi  | P764      | JJK7654 | 00-564    | 12/03/06 | 3           | 40.00    |
|           |           |           |         | 00-770    | 19/06/06 | 2           | 30.00    |

a) Table 2.1 is said to be unnormalized table. Why?

Ans: Some cells have multiple value

b) Transform the unnormalized Table 2.1 into a First Normal Form (1NF) table. Show the transformed 1 NF table. What is the primary key for this new 1 NF table.

Primary Key: StudentID

| StudentID | Name      | LicenseNo | CarReg  | ReceiptNo | Date     | OfficeCode | Fine(RM) |
|-----------|-----------|-----------|---------|-----------|----------|------------|----------|
| F1011     | CK Fong   | P100      | JGK9201 | 00-345    | 10/03/06 | 2          | 30.00    |
| F1011     | CK Fong   | P100      | JGK9201 | 00690     | 15/03/06 | 1          | 20.00    |
| F1011     | CK Fong   | P100      | JGK9201 | 00-810    | 20/03/06 | 2          | 30.00    |
| L782      | AdamJames | P290      | JJL99   | 00-459    | 13/03/06 | 1          | 20.00    |
| H762      | MalaDewi  | P764      | JJK7654 | 00-564    | 12/03/06 | 3          | 40.00    |
| H762      | MalaDewi  | P764      | JJK7654 | 00-770    | 19/03/06 | 2          | 30.00    |

## QUESTION 2

The following table is un-normalized table that is created to manage a consulting company's operations. Create its normalized forms (1NF, 2NF, and 3NF).

Note: Define the PK and FK for each table:

| Project number | Project name           | Employee number | Employee name    | Rate category | Hourly rate |
|----------------|------------------------|-----------------|------------------|---------------|-------------|
| 1023           | Madagascar travel site | 11              | Vincent Radebe   | A             | \$60        |
|                |                        | 12              | Pauline James    | B             | \$50        |
|                |                        | 16              | Charles Ramoraz  | C             | \$40        |
| 1056           | Online estate agency   | 11              | Vincent Radebe   | A             | \$60        |
|                |                        | 17              | Monique Williams | B             | \$50        |

### 1NF:

| Project Number | Project Name           | Employee number | Employee name    | Rate category | Hourly rate |
|----------------|------------------------|-----------------|------------------|---------------|-------------|
| 1023           | Madagascar travel site | 11              | Vincent Radebe   | A             | \$60        |
| 1023           | Madagascar travel site | 12              | Pauline James    | B             | \$50        |
| 1023           | Madagascar travel site | 16              | Charles Ramoraz  | C             | \$40        |
| 1056           | Online estate agency   | 11              | Vincent Radebe   | A             | \$60        |
| 1056           | Online estate agency   | 17              | Monique Williams | B             | \$50        |

### 2NF:

| Project Number | Project Name           |
|----------------|------------------------|
| 1023           | Madagascar travel site |
| 1056           | Online estate agency   |

| Project Number | Employee number | Employee name    | Rate category | Hourly rate |
|----------------|-----------------|------------------|---------------|-------------|
| 1023           | 11              | Vincent Radebe   | A             | \$60        |
| 1023           | 12              | Pauline James    | B             | \$50        |
| 1023           | 16              | Charles Ramoraz  | C             | \$40        |
| 1056           | 11              | Vincent Radebe   | A             | \$60        |
| 1056           | 17              | Monique Williams | B             | \$50        |

**3NF:**

| Project Number | Project Name           |
|----------------|------------------------|
| 1023           | Madagascar travel site |
| 1056           | Online estate agency   |

| Project Number | Employee number | Rate category |
|----------------|-----------------|---------------|
| 1023           | 11              | A             |
| 1023           | 12              | B             |
| 1023           | 16              | C             |
| 1056           | 11              | A             |
| 1056           | 17              | B             |

| Employee number | Employee name    |
|-----------------|------------------|
| 11              | Vincent Radebe   |
| 12              | Pauline James    |
| 16              | Charles Ramoraz  |
| 17              | Monique Williams |

| Rate category | Hourly rate |
|---------------|-------------|
| A             | \$60        |
| B             | \$50        |
| C             | \$40        |